



I. Choose the correct answers:

Q1. Inheritance is a way to

- (A) improve data hiding and encapsulation.
- ☒ (B) add features to existing classes without rewriting them.
- (C) pass arguments to objects of classes.
- ☒ (D) make general classes into more specific classes.

Q2. The dot operator (class member access operator) connects the two entities (left to right):

- ☒ (A) A class object and a member of that class
- (B) A class and a member of that class
- (C) A class member and a class object
- (D) A class object and a class

Q3. A member function can always access the data

- (A) in any object of the class of which it is a member.
- (B) in the public part of its class.
- (C) in the class of which it is a member.
- ☒ (D) in the object of which it is a member.

Q4. Virtual functions allow you to

- (A) create an array of type pointer-to-base class that can hold pointers to derived classes.
- ☒ (B) use the same function call to execute member functions of objects from different classes.
- (C) group objects of different classes so they can all be accessed by the same function code.
- (D) create functions that can never be accessed.

Q5. In a class definition, data or functions designated private are accessible

- (A) only if you know the password.
- ☒ (B) to member functions of that class.
- (C) to any function in the program.
- (D) only to public members of the class.

Q6. An abstract class is useful when

- (A) there are multiple paths from one derived class to another.
- (B) you want to defer the declaration of the class.
- ☒ (C) no objects should be instantiated from it.
- (D) no classes should be derived from it.

Q7. A friend function can be used to

- ☒ (A) increase the versatility of an overloaded operator.
- (B) mediate arguments between classes.
- (C) allow access to classes whose source code is unavailable.
- ☒ (D) allow access to an unrelated class.

Q8. Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- (A) protected members of Derv .
(B) public members of Derv .
(C) private members of Derv .
(D) public members of Base .

**II. Fill in the blanks in each of the following:
(CHOOSE FIVE)**

Q9. In the UML, inheritance is called Generalization

Q10. To be accessed from a member function of the derived class, data or functions in the base class must be public or Protected.

Q11. The only technical difference between structures and classes in C++ is that _____.

Q12. Deciding—after a program starts to execute—what function will be executed by a particular function call statement is called binding or dynamic binding

Q13. Member functions defined inside a class definition are _____ by default.

Q14. A constructor is executed automatically when an object is created (defined)

**III. State whether each of the following is true or false. If false, explain why:
(CHOOSE FOUR)**

Q15. A friend function can access a class's private data without being a member of the class.

Q16. If a base class contains a member function basefunc(), and a derived class does not contain a function with this name, can an object of the derived class access basefunc()? Yes (assuming base func is not private).

Q17. Data items in a class must be private. F (both data and function can be private or public).

Q18. The user must always define the operation of the copy constructor. F (the compiler provides a default copy constructor.)

Q19. In a class you can have more than one constructor with the same name. T

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**IV. Direct questions:
(CHOOSE THREE)**

Q20. What is the purpose of a class definition?

A class declaration describes how object of a class will look when they are created.

Q21. What are the purposes of the this pointer?

Q22. Illustrate the memory representation of static data members.

Q23. Illustrate the virtual base classes by the UML.

V. Write code of the following:

Q24. Write a mini-program that illustrates multiple inheritance.

Q25. Write the Rational class definition using the default parameter values constructor.

```
Class Rational {  
    Public:  
        Rational (int n=0, int d=1): num(n), den(d) {}  
    Private:  
        int num, den;  
};  
void main()  
{  
    Rational x, y(4), z(22, 7);  
    cout << "x = " << endl;
```

```
    x.print();  
    cout << "\ny = " << endl;  
    y.print();  
    cout << "\nz = " << endl;  
    z.print();  
    cout << endl;  
}
```

Q26. Write a class Counter definition code with the static count, and including a constructor for increase the count and a destructor for decrease the count.